

The SSVEP Toolbox: A Python-based, GPU-accelerated toolbox for feature extraction and classification

Daniel Comaduran Marquez¹, Hadi Habibzadeh^{2,3}, Emily Schrag⁴, Eli Kinney-Lang^{1,5}, and James J. S. Norton^{2,3}

¹ Cumming School of Medicine, University of Calgary. Calgary, AB, Canada.

² Research and Development, Stratton VA Medical Center, Albany, NY, United States of America.

³ Electrical and Computer Engineering, University at Albany, Albany, NY, United States of America.

⁴ Neurosciences, University of Calgary. Calgary, AB, Canada.

⁵ Biomedical Engineering, University of Calgary. Calgary, AB, Canada.

E-mail: dcomadur@ucalgary.ca

Abstract. We present the steady-state visual evoked potential (SSVEP) Toolbox, an open-source; Python-based; multi-threaded; and graphical processing unit (GPU)-accelerated data analysis toolbox for SSVEP feature extraction and classification. The toolbox provides easy-to-use implementations of three feature extraction methods for data classification: filter bank canonical correlation analysis, minimum energy combination, and multivariate synchronization index. The SSVEP Toolbox was validated through two classification analyses and profiling of the runtime in different CPU and GPU systems. The datasets used for classification were: 1) the Wang2016 from the Mother of All BCI Benchmarks (MOABB) and 2) a pediatric dataset acquired at the BCI4Kids lab in the Alberta Children's Hospital. The classification analyses using the three feature extraction methods quantified the accuracy of the classifiers in the toolbox compared to a Riemannian geometry with logistic regression classifier. Profiling of the toolbox was done using the pediatric dataset by comparing runtimes using four different CPU and GPU systems. Classification accuracies were comparable to the RG with logistic regression classifier and the runtime of GPU implementations (0.29 ± 0.01 sec, mean $\pm$ SD) outperformed the CPU (3.69 ± 0.17 sec, mean $\pm$ SD) ones. The SSVEP Toolbox is a suitable solution for SSVEP-BCI applications that is written in a free and open-source programming language (i.e., Python). Thus, making it more accessible not only to the research community but also end-users to be implemented in real-life applications. Future work should focus on validating the toolbox in online experiments and extending its features to include more feature extraction methods and compatibility with non-Compute Unified Device Architecture (CUDA) GPUs.

Keywords: Brain-computer interface, SSVEP feature extraction, SSVEP classification

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1. Introduction

The field of brain-computer interfaces (BCIs) has seen a growing interest in the use of steady-state visual evoked potentials (SSVEPs) as a control paradigm [1]. SSVEPs are electroencephalography (EEG) responses, usually recorded over the occipital region of the scalp, to visual stimulation at a fixed frequency [2]. SSVEP-based BCIs have multiple advantages compared with other BCI paradigms; for example, they don't require extensive user training [3] and may require less cognitive effort [4]. However, the high-contrast visual stimuli commonly used in SSVEP can also lead to increased visual fatigue or discomfort [5].

Most BCI research is focused on addressing problems for adult populations, particularly those with paralysis or neurodegenerative diseases (e.g., Amyotrophic Lateral Sclerosis). However, pediatric populations could also greatly benefit from BCI technologies. For example, cerebral palsy (CP) is the most common cause of physical disability in childhood [6]. Previous research has shown that children with CP can use BCI-based neurofeedback rehabilitation to encourage early neuro-plastic changes which could improve motor control [7]. The high neuroplasticity and lifelong needs of pediatric populations may mean they have the ability and motivation to be lifelong BCI users. Previous research has shown that healthy children [8], and children with disabilities [9–14], can use BCIs based on motor imagery, mental commands, event-related potentials (e.g., P300), and SSVEPs for multiple different applications.

Recent advancements in SSVEP-based BCIs have focused on improving the robustness and accessibility (i.e., written in an free and open-source programming language) of the underlying feature extraction and classification algorithms. One key development has been the introduction of open-source BCI frameworks like the Mother of All BCI Benchmarks (MOABB) [15], which provide a standardized platform for evaluating and comparing different BCI algorithms. However, MOABB does not provide native graphical processing unit (GPU) implementation, which can be crucial for real-time applications. Additionally, many novel SSVEP classifiers are still primarily developed in scientific programming languages like MATLAB, which can make them inaccessible to some researchers and end-users, or create other barriers to their implementations in BCI systems. Thus, there is a need for open-source, GPU-accelerated SSVEP Toolboxes, written in a freely available, widely-used language to facilitate community contributions and improve the robustness of these tools.

To address these limitations, this project developed the SSVEP Toolbox. For SSVEP feature extraction, the toolbox provides easy-to-use implementations of three state-of-the-art feature extraction methods: (1) filter bank canonical correlation analysis (CCA) [16], (2) minimum energy combination (MEC) [17], and (3) multivariate synchronization index (MSI) [18]. The SSVEP Toolbox allows users to perform classification using a single feature extraction method or by combining multiple feature

extraction methods through voting [19]. To enhance the accessibility and performance of the SSVEP Toolbox, it is open-source (Python-based); multi-threaded; and GPU-accelerated. We validated the toolbox by performing two classification analyses and profiling its performance. Each classification analysis used a dataset—Wang2016 dataset from MOABB, as well as data collected from typically developed children, ages 12.3 ± 4.2 (mean $\pm$ SD)—to quantify the accuracy of classifiers instantiated using the SSVEP Toolbox compared to a classifier based on Riemannian geometry (RG) with logistic regression.

2. Methods

The SSVEP Toolbox was implemented using Python (version 3.11) [20] and presently includes three training-free feature extraction methods: CCA [21], MEC [17], and MSI [18]. Classification can be performed based on any of the three feature extraction methods (using a threshold) or using multiple feature extractions methods and voting. To enhance performance, the toolbox can perform feature extraction and classification of multichannel EEG data using the CPU (multithreaded; default) or a Compute Unified Device Architecture (CUDA)-enabled GPU.

First, we will describe the basic input and output objects, and parameters used by the SSVEP toolbox. Then, we will describe the basis of the three training-free feature extraction methods included in the toolbox. In the equations defined below, italicized lowercase letters (e.g., x) refer to scalar values, italicized lowercase letters with an arrow (e.g., $\vec{a}$) refer to vectors, and bold and italicized letters (e.g., $\mathbf{A}$) refer to matrices.

2.1. Toolbox settings

2.1.1. Input

The SSVEP toolbox takes a three-dimensional array with dimensions: signals, electrodes, and samples, where signals is the number of signals or epochs. If the toolbox is set to be used in online analysis, this dimension should be of length = 1. Electrodes and samples are the number of EEG electrodes and number of EEG data samples per signal or epoch, respectively.

2.1.2. Parameters

The mandatory and optional parameters for the three feature extraction methods are described below:

- Mandatory parameters:
 - Harmonics: Number of harmonics to be used in the template signals. Should be a positive integer between 3 and 5.

- Target frequencies: Stimulation frequencies of each target in Hz. The length of this variable determines the length of the target dimension in the output. The number of target frequencies depends on the experimental setup.
- Sampling frequency: Sampling frequency of the EEG signal in Hz. It must be a real positive value.
- Optional parameters:
 - Embedding dimension: Dimension of the time-delay embedding. This must be a positive integer. The default is zero, thus, no time embedding is used. If there are E EEG electrodes and we set the embedding dimension to n , the input signal will be expanded to have $E \times n$ channels. The length of the samples is truncated to retain the original length.
 - Delay step: Number of samples that are shifted for each embedding dimension. This must be a positive integer.
 - Filter order: The n-order of the filter used for pre-processing. If this variable is set to zero, no filtering is performed. This parameter must be a positive integer. The default is zero.
 - Low cut-off frequency: Lower cut-off frequency of the pre-processing filter in Hz. This must be a single positive number. The default is zero, so no filtering is performed.
 - High cut-off frequency: Higher cut-off frequency of the pre-processing filter in Hz. This must be a single positive number. The default is zero, so no filtering is performed.
 - Sub-bands: Sub-band values used to create filter-bank. Must be a two column matrix where each row defines a sub-band in Hz. The values in the columns should be the low and high cut-off frequencies, respectively. The default is none, so no sub-band filtering is done.
 - Voters count: Number of electrode-selections used for feature extraction. This must be a positive integer. The default is set to 1, thus, all electrodes are used. If set to a positive integer other than 1, the feature extraction will be performed in n-randomized combinations of electrodes (i.e., voters).
 - Use GPU: Boolean to extract features using a CUDA-enabled GPU. Default is set to false, thus, using the CPU.
 - Explicit multithreading: If set to n, the feature extraction will use n-multiple threads to process signals/voters in parallel. Can only be used when doing feature extraction in CPU-mode. The max value of n is limited by the CPU. The default is set to 1, so no multithreading is used.
 - Max batch size: Maximum number of signals/channels selections that are processed in one batch. This must be a single positive integer. Increasing the value of this number may improve parallelization at the expense of memory usage. The default is set to none, so all signals are processed in a single batch.
 - Samples count: If the number of samples is provided, the toolbox will pre-compute the template signal with the same length as samples count. This must

be a positive integer. Pre-computing the template signal saves on computing and memory resources. This parameter should be the same as the samples length in the input signal.

2.1.3. Output

The resulting feature extraction is stored in a five-dimensional array with dimensions: signals, sub-bands, voters, targets, features. The signals dimension will be the same length as the input signal dimension previously described. The second dimension indexes sub-bands with the same length as the sub-bands parameter. If no sub-bands were provided, the sub-band dimension will be of length = 1. The third dimension indexes voters (i.e., channel selection). This dimension will be the same length as the number provided in the voters parameter. The fourth dimension indexes the target frequencies. The last dimension indexes the number of features per signal or epoch.

2.2. Feature extraction methods

2.2.1. Canonical Correlation Analysis

The CCA feature extraction is based on the implementation described by Chen et al. [16]. CCA is a statistical method used to measure the linear correlation between two multi-dimensional variables. Consider two multi-dimensional variables $\mathbf{X}$, and $\mathbf{Y}$ with linear combinations $\mathbf{x} = \mathbf{X}^T \mathbf{W}_x$ and $\mathbf{y} = \mathbf{Y}^T \mathbf{W}_y$, respectively. CCA finds the weight vectors $\mathbf{W}_x$ and $\mathbf{W}_y$ that maximize the correlation between $\mathbf{x}$ and $\mathbf{y}$ by solving Eq. 1

$$\max_{\mathbf{W}_X, \mathbf{W}_Y} \rho(x, y) = \frac{E[\mathbf{W}_X^T \mathbf{X} \mathbf{Y}^T \mathbf{W}_Y]}{\sqrt{E[\mathbf{W}_X^T \mathbf{X} \mathbf{X}^T \mathbf{W}_X] E[\mathbf{W}_Y^T \mathbf{Y} \mathbf{Y}^T \mathbf{W}_Y]}} \quad (1)$$

Where the highest ρ is the maximum canonical correlation. The $\mathbf{X}$ signal is the multi-channel EEG signal; and the $\mathbf{Y}$ signal is a reference signal. The reference signal is made from sinusoidal signals with the same frequency as the SSVEP stimuli and its N-harmonics (Eq. 2). The maximum canonical correlation can be computed between the multi-channel EEG signal and reference signals at the frequencies of interest for SSVEP classification.

$$\mathbf{Y}_f = \begin{bmatrix} \sin(2\pi f t) \\ \cos(2\pi f t) \\ \vdots \\ \sin(2\pi N_h f t) \\ \cos(2\pi N_h f t) \end{bmatrix} \quad (2)$$

2.2.2. Minimum Energy Combination

The MEC classifier is based on the implementation described by by Friman et al. [17]. MEC combines EEG signals to cancel as much of the nuisance (i.e., non-SSVEP

response) signals as possible. The measured SSVEP response from the EEG signal can be modeled as Eq. 3.

$$Y = XA + ZB + E \quad (3)$$

Where $\mathbf{Y}$ is a $N_s \times N_c$ matrix with the EEG signal of dimensions [samples $\times$ channels], $\mathbf{E}$ is a noise matrix with the same dimensions as $\mathbf{Y}$, and $\mathbf{A}$ and $\mathbf{B}$ contain the amplitudes and scaling factors for all sinusoids ($\mathbf{X}$) and nuisance signals ($\mathbf{Z}$), respectively. To use MEC for feature extraction, first, the potential SSVEP components are removed from the EEG by projecting them onto the orthogonal complement of an SSVEP model matrix (Eq. 4).

$$X = \begin{bmatrix} X_1 & X_2 & \dots & X_{N_h} \end{bmatrix} \quad (4)$$

Where each submatrix $\mathbf{X}_k$ contains a $\sin(2\pi kft)$ and $\cos(2\pi kft)$ pair of columns with the corresponding f stimulus frequencies. Following this, the matrix ($\tilde{Y}$) will contain only nuisance signals and noise (i.e., $\tilde{Y} = ZB + E$). Next, we find a weight vector ($\vec{w}$) to minimize the resulting energy of the combination of electrode signals (Eq. 5).

$$\min_{\vec{w}} \|\tilde{Y}\vec{w}\|^2 = \min_{\vec{w}} \vec{w}^T \tilde{Y}^T \tilde{Y} \vec{w} \quad (5)$$

The solution to the minimization problem is given by the smallest eigenvector $\mathbf{v}_1$, and thus, the energy of the resulting combination equals the smallest eigenvalue λ_1 . The number of channels is then reduced by keeping only the number of channels that satisfy the relationship shown in 6. Thus, giving us an estimate of the sinusoids ($\mathbf{X}$) by discarding as close as 90% of the nuisance signal and noise ($\mathbf{ZB} + \mathbf{E}$) energy as possible.

$$\frac{\sum_{i=1}^{N_s} \lambda_i}{\sum_{j=1}^{N_s} \lambda_j} > 0.1 \quad (6)$$

A specific SSVEP response can be calculated (Eq. 7).

$$T = \frac{1}{N_s N_h} \sum_{l=1}^{N_s} \sum_{k=1}^{N_h} \frac{\hat{P}_{k,l}}{\hat{\sigma}_{k,l}^2} \quad (7)$$

Where $\hat{P}_{k,l}$ is the estimated power in the k^{th} SSVEP harmonic frequency for the l^{th} channel, and $\hat{\sigma}_{k,l}^2$ is an estimate of the noise power at the same frequency. The signal power in the k^{th} SSVEP harmonic in the channel s_l can be calculated as shown in Eq. 8. Where $\mathbf{X}$ is the reconstructed EEG signal keeping the eigenvalues described in Eq. 6.

$$\hat{P}_{k,l} = \|\mathbf{X}_k^T s_l\|^2 \quad (8)$$

The noise power estimates ($\hat{\sigma}_{k,l}^2$) represent the power in the SSVEP frequencies if no SSVEP response would be present. Friman et al. [17] have chosen an autoregressive model fitted by invoking the Wiener-Khinchin theorem for computing the auto-covariance of each channel signal and then solving the Yule-Walker equations using a Levinson-Durbin recursion. The noise level can be estimated as shown in Eq. 9.

$$\hat{\sigma}_{k,l}^2 = \frac{\pi N_t}{4} \frac{\hat{\sigma}^2}{\left| 1 + \sum_{j=1}^p \exp(2\pi i j k f / F_s) \right|^2} \quad (9)$$

Where N_t is the number of samples, f is the SSVEP stimulation frequency, F_s is the sampling frequency, and i is the complex number $\sqrt{-1}$.

2.2.3. Multivariate Synchronization Index

MSI measures the synchronization between a signal of interest and a reference signal as described by Zhang et al. [18]. Similar to CCA, a reference signal is constructed (Eq. 2). Both the signal of interest ($\mathbf{X}$), and the reference signal ($\mathbf{Y}$) are normalized to have zero mean and unitary variance. Then, a covariance matrix is computed as follows:

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} \quad (10a)$$

$$C_{11} = \frac{1}{M} \mathbf{X} \mathbf{X}^T \quad (10b)$$

$$C_{22} = \frac{1}{M} \mathbf{Y} \mathbf{Y}^T \quad (10c)$$

$$C_{12} = C_{21} = \frac{1}{M} \mathbf{X} \mathbf{Y}^T \quad (10d)$$

Where M is the number of samples in the signals. To reduce the influence of the autocorrelation of $\mathbf{X}$ (Eq. 10b) and $\mathbf{Y}$ (Eq. 10c) the following linear transformation is used:

$$U = \begin{pmatrix} C_{11}^{-1/2} & 0 \\ 0 & C_{22}^{-1/2} \end{pmatrix} \quad (11)$$

A new correlation matrix is computed (Eq. 12). The normalized eigenvalues can then be used to compute the synchronization index (β_k) (Eq. 13).

$$C' = U C U^T \quad (12)$$

$$\beta_k = 1 + \frac{\sum_{i=1}^p \lambda'_i \log(\lambda'_i)}{\log(p)} \quad (13)$$

Where λ' are the normalized eigenvalues, $p = N + 2N_H$, N is the number of channels, N_H the number of harmonics, and k is the index of K^{th} reference signals. Thus, the target frequency will be the one with the highest β_k .

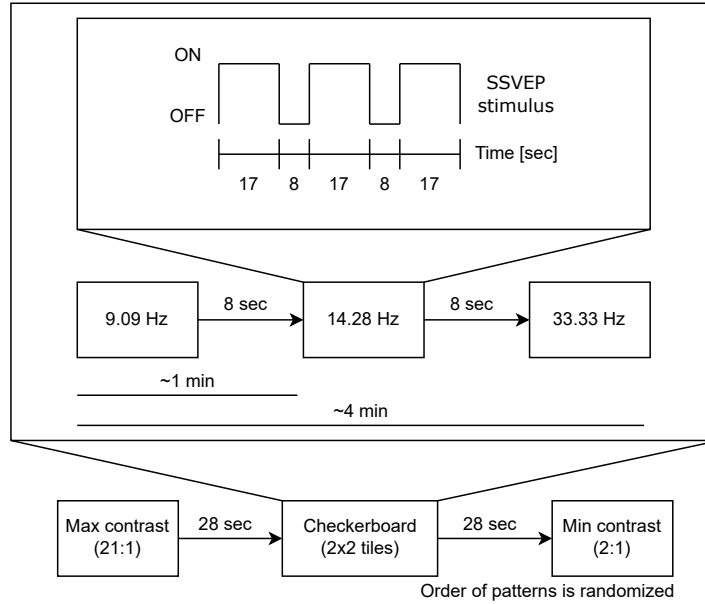


Figure 1: Diagram of timing presentation of the SSVEP stimuli. Stimuli flashed against a black background. SSVEP stimuli are shown for 17 sec, followed by a 8 sec period showing only the background. This was repeated 3 times, following a 28 sec resting period where the subject could rest their eyes. This stimulation and resting periods were repeated for each frequency and pattern (i.e., max contrast, checkerboard, and min contrast). The order of the patterns was randomized to avoid biasing effects.

2.3. Datasets

Two datasets were used to validate the SSVEP Toolbox:

2.3.1. Wang 2016 - Benchmark Dataset for SSVEP-Based BCIs

The first dataset was the Wang2016 dataset [22] taken from the MOABB tool [15]. The dataset includes recordings from 35 healthy subjects (17 females, aged 17-34 years, mean age: 22 years). The 64-channel EEG data was acquired with a Synamps2 system (Neuroscan, Inc., Abbottsford, Australia) with a sampling rate of 1000 Hz, downsampled for analysis to 250 Hz. The amplifier frequency passband ranged from 0.15 Hz to 200 Hz. Frequency stimuli go from 8 to 15.8 Hz with 0.2 Hz intervals (i.e., 40 stimulation frequencies). For each subject, the experiment consisted of 6 blocks. Each block contains 40 trials (i.e., one for each stimulation frequency) that appeared in a random order. Each trial started with a visual cue shown for 500 msec to indicate the target stimulus. Following the cue, all stimuli were active concurrently for 5 sec. After the stimulus period there was a blank screen for 500 msec. Thus, each trial lasted a total of 6 sec. This dataset was chosen as it has a high number of targets, is well-known, and has been used previously for bench-marking and validation purposes [23].

2.3.2. BCI4Kids lab 2024 - Typically-developing pediatric population

The second dataset was obtained from a normally-developing pediatric population ($n=35$, 20 females) with ages = 12.3 ± 4.2 (mean $\pm$ SD), and is a subset of data from an additional ongoing study at the BCI4Kids Lab at the Alberta Children’s Hospital. The study was approved by the Conjoint Health Research Ethics Board (CHREB) of the University of Calgary (REB23-1232). All participants (or their parent or legal guardian in the case of children under 16) gave written informed consent to participate in the study. EEG data was recorded with a sampling rate of 256 Hz using the gel-based g.tec g.GAMMAsys system, g.USBamp amplifier, and g.GAMMAcap headcap (g.tec, Schiedlberg, Austria). A montage of 16 electrodes (Oz, POz, Pz, Cz, O1, O2, PO7, PO8, P3, P4, P7, P8, T7, T8, C3, C4) was placed in accordance with standard 10-20 system locations, with a common reference electrode at FPz. The stimuli used to generate the dataset consist of three different patterns: (1) a maximum contrast (21:1) pattern, (2) a checkerboard pattern with 2×2 tiles, and (3) a minimum contrast (2:1) pattern (Fig. 1). The stimulus presentation paradigm was designed using an BCI framework built in Unity [24]. All stimuli were presented on a 32” LG monitor with a refresh rate of 100 Hz. The order of the patterns was randomized for each participant to avoid biasing effects. Frequency values of 9.09, 14.28 and 33.3 Hz were chosen to cover the whole frequency range of SSVEP stimuli and to comply with the refresh rate of the presentation monitor. The size of the stimulus was approximately 15.87 cm^2 . The stimuli flashed against the solid black background for 17 seconds, followed by a period of 8 seconds where only the background was visible. This pattern was repeated 3 times. After the 3 stimulus periods where finished, there was 28 second rest period where the subjects could rest their eyes without having to look at the screen. The stimulation and resting periods where repeated for each frequency. This protocol was then repeated for each pattern (i.e., max contrast, checkerboard, and min contrast). The order of the patterns was randomized to avoid any biasing effects.

2.4. Data processing pipeline

2.4.1. Pre-processing

Both datasets were filtered using a 4th order Butterworth filter with cut-off frequencies (f_c) of 0.1 and 100 Hz, respectively. Additionally, a 4th order IIR notch filter at 60 Hz was implemented. The filtered signals were then trimmed. The MOABB dataset signals were trimmed into 5-second segments according to the event IDs provided in the dataset. The pediatric dataset signals were trimmed into 17-second segments according to the SSVEP-ON markers reported by Unity.

Given the visual nature of the SSVEP paradigm, only occipital channels (i.e., O1, Oz, and O2) were considered for feature extraction and classification, with other channels ignored.

2.4.2. Feature extraction and classification

The feature extraction settings for the MOABB dataset were as follows. All feature extraction methods used the target frequencies from 8 to 15.8 Hz with an interval of 0.2 Hz with 3 harmonics:

- fbCCA: Sub-bands with low cut-off frequency from 8 to 80 Hz in 8 Hz increments, and high cut-off frequency of 88 Hz with a 12th order Butterworth band-pass filter.
- MEC: Auto-regression order of 15 with energy ratio of 0.05 (i.e., to keep 95% of the signal of interest).
- RG with logistic regression: Ledoit-Wolf shrunk covariance matrix estimator with large-scale bound-constrained optimization linear regression.

The feature extraction settings for the pediatric dataset were the same as the MOABB dataset with the exception of the filter-bank used for the fbCCA method. For the pediatric dataset, the filter-bank had low cut-off frequencies from 1 to 37 Hz in 3 Hz increments, and a high cut-off frequency of 40 Hz with a 12th order Butterworth band-pass filter.

Note that MSI does not have any hyper-parameter settings to be tuned. The classification for fbCCA, MEC, and MSI simply used the index of the maximum argument (i.e., CCA, signal-to-noise ratio, and synchronization index, respectively) corresponding to the predicted target frequency.

To validate the SSVEP Toolbox using both datasets, classification accuracy for targets (i.e., 40 targets per trial for dataset 1, and 3 targets per trial for dataset 2) were computed for all the classifiers in the toolbox. As an additional validation, a filter-bank Riemmanian geometry with logistic regression classifier (as detailed in MOABB) was also compared to the implemented toolbox as a gold-standard reference [25]. The classification accuracy for each subject was plotted as a box plot with each individual subject represented as an overlaid dot in the box plots.

2.4.3. Profiling

To test the performance of the tool for offline processing with the pediatric dataset, the SSVEP Toolbox classifiers were profiled in their CPU and GPU implementations using 1 and 10 voters. The profiler code does 5 warm-up trials (i.e., 5 runs where no timing data is recorded). These warm-up trials are used to reduce timing fluctuation and exclude the overhead in first invocations of a function. After the warm-up trials, the mean time $\pm$ SD of 5 runs is used to profile the performance of the classifiers. The following systems were used in the profiling analysis:

- AMD Ryzen 7 5800 HS with 16 GB of DDR4 RAM.
- Intel i7 6700 with 48 GB of DDR4 RAM
- NVIDIA GeForce RTX 3050 Laptop GPU with 4 GB of VRAM
- NVIDIA RXT A5000 with 24 GB of VRAM

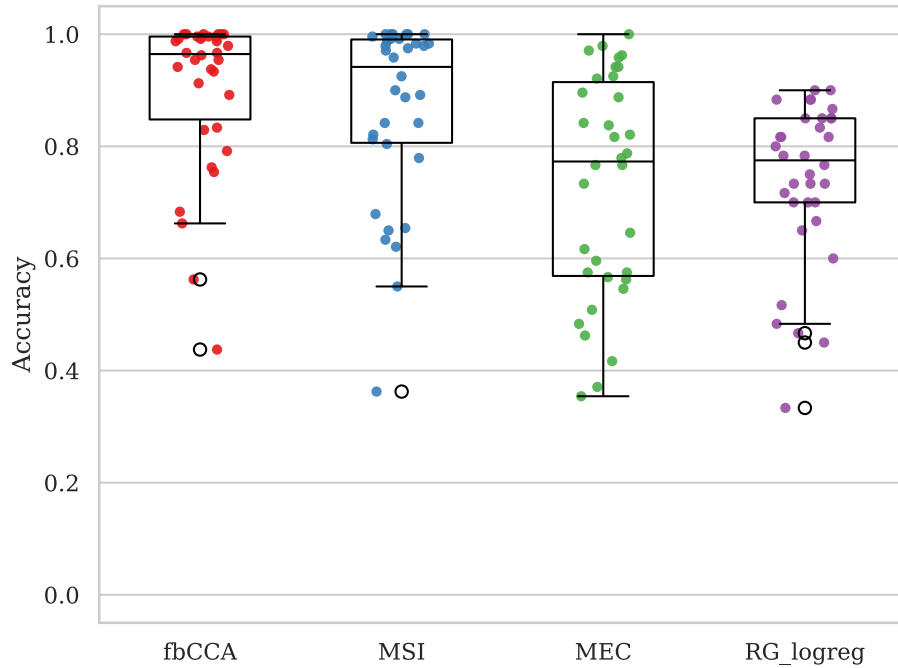


Figure 2: Classification accuracy box plot of the Wang2016 dataset. The boxes extends from the first (Q1) to the third (Q3) quartile, The whiskers extend from the box to the farthest data point lying within 1.5x the inter-quartile range (IQR) from the box, the median is represented by the line inside the box, and the small circles represent outliers. Each colored dot represents the accuracy for an individual subject.

3. Results

3.1. Classification accuracy

The classification accuracy using the Wang2016 dataset shows high accuracy for all classifiers (i.e., higher than 70%) (Fig. 2). The classifiers we implemented that use fbCCA and MSI for feature extraction had higher median accuracies than the classifiers we implemented using MEC and Riemaniann geometry.

The classification accuracy using the pediatric dataset shows that all the classifiers have high accuracy (i.e., higher than 70%) (Fig. 3). Conversely to the Wang2016 dataset, the pediatric dataset shows the highest accuracy for the fbCCA and Riemaniann geometry classifiers.

3.2. SSVEP tool profiling

The profiling shows that the MSI and CCA classifiers performed slightly faster than MEC in both the single- and multiple-voter modalities. Additionally, the GPU results show shorter times across the classifiers for both voter modalities, with the highest improvement (i.e., shorter times) being for the MSI classifier with the

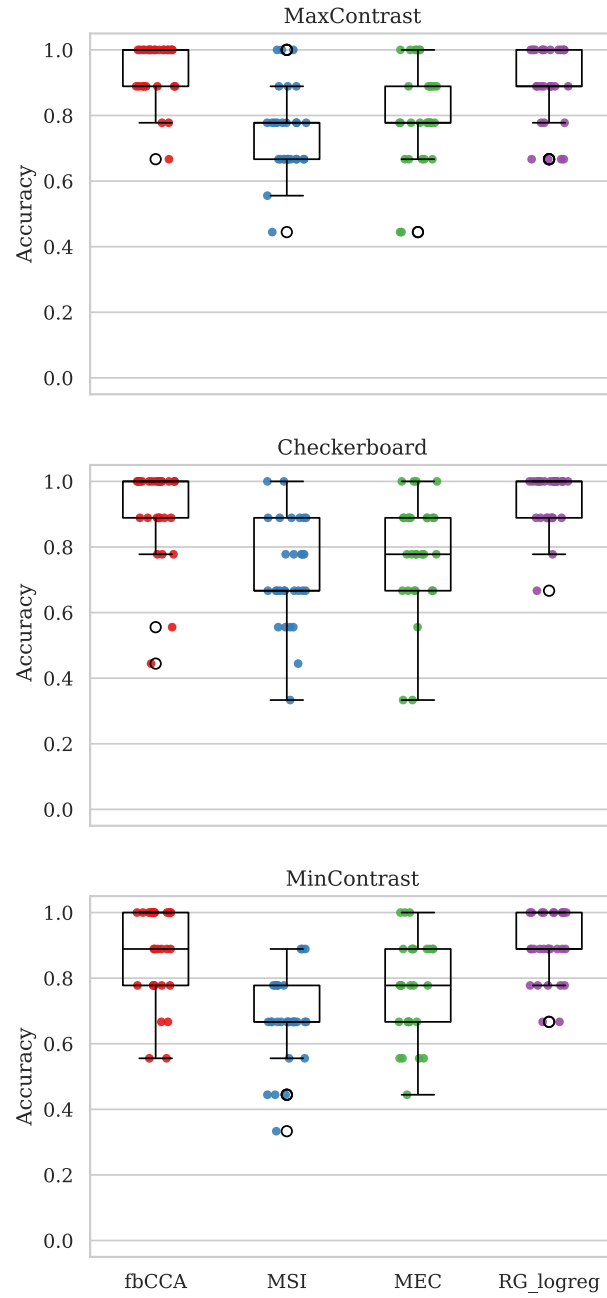


Figure 3: Classification accuracy box plot of the pediatric dataset per stimulus texture. The boxes extends from the first (Q1) to the third (Q3) quartile, The whiskers extend from the box to the farthest data point lying within 1.5x the inter-quartile range (IQR) from the box, the median is represented by the line inside the box, and the small circles represent outliers. Each colored dot represents the accuracy for an individual subject.

NVIDIA GeForce RTX 3050 graphics processor (Fig. 4). In average, the GPU (0.29 ± 0.01 sec, mean $\pm$ SD) outperformed the CPU (3.69 ± 0.17 sec, mean $\pm$ SD) implementations.

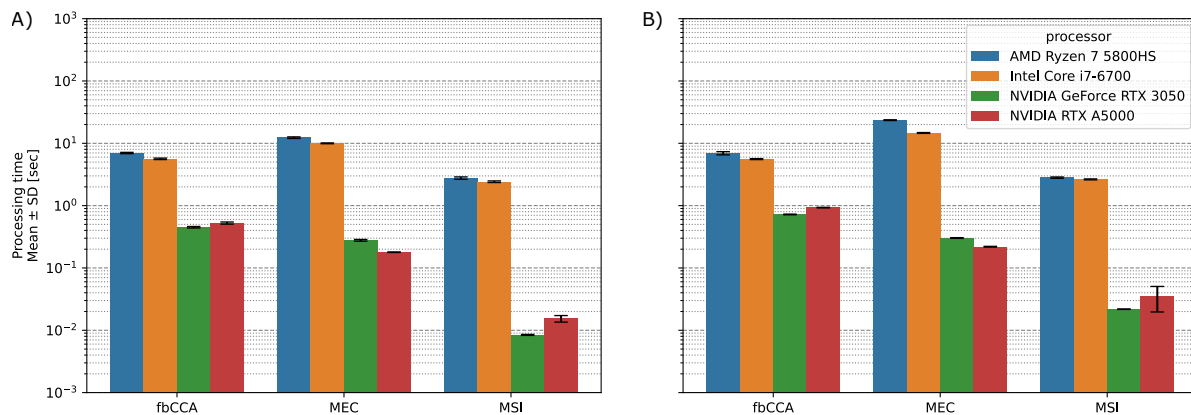


Figure 4: A) Processing time (mean $\pm$ SD) of the SSVEP Toolbox classifiers using an Intel Core i7 6700, an AMD Ryzen 7 5800HS, and NVIDIA RTX 3050 and RTX A5400 GPUs using one subject from dataset 2 using a single voter (A) and 10 randomized voters (B). Lower bars (i.e., shorter times) are better. The profiling tool was ran 5 times with 5 warm-up trials.

4. Discussion

4.1. SSVEP classifier validation

The proposed SSVEP Toolbox, which includes implementations of CCA, MEC, and MSI feature extraction methods, demonstrated classification accuracy similar to the previously reported literature [16]. When comparing the accuracy of the classifier using the Wang2016 dataset, fbCCA and MSI had higher accuracy than the MEC and Riemaniann geometry (Fig. 2). Conversely, in the pediatric dataset the classifier with highest accuracy are fbCCA and Riemaniann geometry classifiers (Fig. 3). Nevertheless, for both datasets, the classifiers in the SSVEP tool box show accuracies over 70%. The 70% limit has been previously using by the BCI community to distinguish between high and low BCI performances [26]. These high accuracies suggest that the toolbox’s implementations are reliable and consistent with existing methods.

There are two key features of the SSVEP toolbox: 1) it is implemented in a open-source and accessible programming language (i.e., Python), and 2) it has integration with CUDA-enabled GPU systems. The first feature promotes transparency, reproducibility, and collaboration within the research community, as well as facilitating the integration of the toolbox into various applications and workflows. The second feature allows for faster feature extraction and classification by leveraging the CUDA-enabled implementation. However, further development should enable compatibility with non-CUDA-enabled GPUs (e.g., AMD or Intel-based GPUs). This would broaden the toolbox’s accessibility and allow for more diverse hardware configurations to be utilized. Additionally, the toolbox’s performance could be further optimized by incorporating tools such as JAX, which can pre-compile matrix computations and potentially improve the speed of the feature extraction and classification processes. This

could be particularly beneficial for real-time applications, such as BCI applications for children.

Offline profiling of the SSVEP Toolbox with the pediatric dataset shows reduced processing times when using the GPUs vs the CPUs (Fig. 4). To further validate the SSVEP Toolbox and evaluate its performance in different contexts, future work should involve conducting online experiments. To validate the use of the voting system, online experiments should include testing with individuals with atypical brain anatomy, allowing the SSVEP Toolbox to use only the information from channels over unaffected brain regions. This would provide valuable insights into the toolbox’s robustness and potential limitations, as well as inform any necessary adjustments or enhancements to the algorithms or implementations.

Finally, the multichannel voting system implemented in the SSVEP Toolbox, which creates randomized channel subsets for feature extraction, could be further explored and compared to other channel selection strategies [19]. This analysis could shed light on the optimal approaches for leveraging channel information and potentially improving classification accuracy and reliability.

5. Conclusion

We have presented a new open-source, Python-based, multi-threaded, and GPU-accelerated data analysis toolbox for SSVEP feature extraction and classification. The toolbox provides easy-to-use implementations of three state-of-the-art feature extraction methods: filter bank CCA, MEC, and MSI. Additionally, the SSVEP Toolbox includes a voting system to create randomized channel subsets, which can improve feature extraction performance. The toolbox was validated using publicly available dataset (i.e., Wang2016), as well as with a dataset from a normally developing pediatric population. Future work should focus on improving the performance of the tool, and expanding it to be able to work with non-CUDA GPUs. Additionally, online experiment validation, as well as validation with individuals with abnormal brain anatomy could be useful to test the channel selection capabilities.

6. Data availability statement

The data cannot be made publicly available upon publication because they contain sensitive personal information. The data that support the findings of this study are available upon reasonable request from the authors.

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8. References

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